

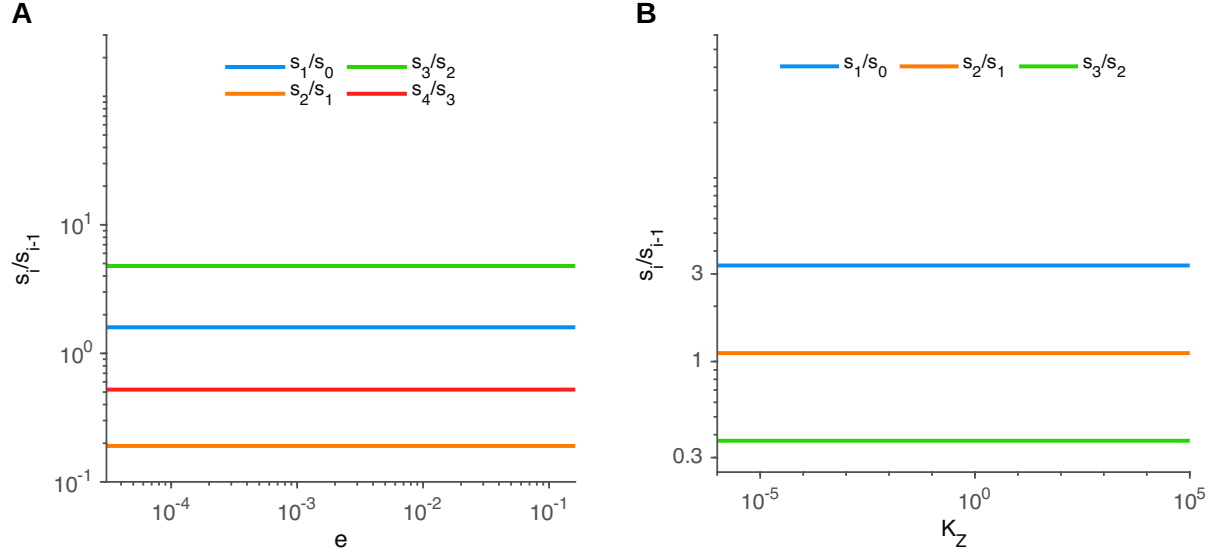


Figure S1: Numerical checks of the steady-state invariant. (A) Adjacent free-target ratios over a 5,278-fold enzyme sweep at fixed free effector in a four-site nonidentical ladder. (B) A three-site example with a reversible dead end swept over eleven decades at fixed total effector. The small ratio drift reflects the changed free effector; it is not a violation of the free-input identity. See S10 and the source parameter records.